

Chapter 1

Integers and Congruence

1.1 Integers

Much of abstract algebra studies the set of integers, subsets of integers, or sets which are equivalent to subsets of integers. Thus to move forward, it will behoove us to step back and investigate the properties of integers. You have been using many of these properties all of your life. Taking a rigorous look at these properties will not only enrich your understanding, but will allow you to apply these fundamental properties to more general sets.

Definition 1. The Well Ordering Principle. Every nonempty set of positive integers contains a smallest member.

We will treat the well ordering principle as an axiom.¹

Definition 2. We say that a nonzero integer d *divides* an integer m (or say that d is a *divisor of* m) if m can be written as, $m = dn$ for some integer n , denoted $d|m$.

- $d|m \Rightarrow m = dn$ for some integer n .
- $d \nmid m \Rightarrow m \neq dn$ for any integer n .

Example 1. $3|12$ since $12 = 3 \cdot 4$ (which also shows that $4|12$). $3 \nmid 10$ since there is no integer n such that $3 \cdot n = 10$ (solving for n , $n = 10/3 \notin \mathbb{Z}$).

Exercise 1. Prove that If $a|b$ and $b|c$ then $a|c$.

Exercise 2. If $a|b$ and $a|c$ then $a|b \pm c$.

Definition 3. A prime p is an integer greater than 1 that has no divisors other than 1 and p .

- Equivalently, p is prime if and only if $d|p \Rightarrow d = 1$ or $d = p$.

Example 2. 5 is prime since $1 < 5$ and $2, 3, 4 \nmid 5$.

Exercise 3. ★ Prove that there are infinitely many prime numbers.

Theorem 1. Division Algorithm. Let $a, b \in \mathbb{Z}$ and $a, b > 0$ then there exists a unique $q, r \in \mathbb{Z}$ such that $a = bq + r$ where $0 \leq r < b$.

- r is called the remainder upon dividing a by b .

Example 3. If you were to divide 13 by 3, the division algorithm tells us that $13 = 3 \cdot 4 + 1$; where $a = 13$, $b = 3$, $q = 4$, and $r = 1$. Once we have chosen a and b (or q), the theorem tells us that a can be written uniquely in this form.

- Note: we can write 13 as $13 = 4 \cdot 2 + 5$, $13 = 4 \cdot 1 + 9$, or $13 = 4 \cdot 5 - 7$. However the remainders are 5, 9 $>$ 4 in the former cases and, $-7 <$ 0 in the latter case. The assumption that $0 \leq r < b$ is what guarantees uniqueness.

Example 4. $57 = 12 \cdot 4 + 9$.

¹Meaning we will just assume it to be true for the integers. The well ordering principle sometimes given as a theorem proved with an induction (see below). As it turns out, induction and the well ordering principle are logically equivalent, and we will instead show that induction follows from the well ordering principle.

Proof. There are two parts to prove, existence and uniqueness. First we show that you can write any integer a as $bq + r$ where $0 \leq r < b$. Let $S = \{a - bq : q \in \mathbb{Z} \text{ and } a - bq \geq 0\}$. First note that S is nonempty. If $a - b < 0$ then let $k = -1$ so that $a - (-1)b > 0$. If $a - b \geq 0$ then let $k = 1$.

If $0 \in S$ then $a = bk$ which implies that $b|a$ and then we can let $q = a/b$ (an integer since $b|a$) and $r = 0$ so that $a = b(a/b) + 0$. If $0 \notin S$ then we can apply the well ordering principle (since S is nonempty) to choose the smallest number in S , say $r = a - bq$. Then $a = bq + r$; $r \geq 0$ per our definition of S so we need only show that $r < b$. If $r \geq b$ then $r - b \geq 0 \Rightarrow a - bq - b = a - b(q + 1)$ so that $a - b(q + 1) \in S$. But then $a - bq > a - b(q + 1)$ and we already assumed that $a - bq = r$ was the *smallest* number in S – a contradiction. So $r < b$.

Now we show uniqueness. Let $a = bq + r$ and $a = bq' + r'$ such that $0 \leq r < b$ and $0 \leq r' < b$ for integers q, q', r, r' . Without loss of generality² we can assume that $r' \geq r$. $bq + r = bq' + r'$ and $bq' + r' - (bq + r) = 0 \Rightarrow b(q' - q) = r' - r$. Thus $b|(r' - r)$ and $0 \leq r' - r \leq r' < b$. The only way b can divide a number smaller than itself (and still be an integer) is if that number is 0. So then $r' - r = 0 \Rightarrow r = r'$ which also shows that $q' = q$. $\square$

Definition 4. The *greatest common divisor* (or GCD) of integers a and b is the largest integer that divides both a and b ; denoted $\gcd(a, b)$. When $\gcd(a, b) = 1$ we say that a and b are *relatively prime*.

Example 5. $\gcd(10, 6) = 2$ since $10 = 2 \cdot 5$ and $6 = 2 \cdot 3$.

Example 6. $1100 = 2^2 5^2 11$ and $440 = 2^3 5 \cdot 11$. So $\gcd(1100, 440) = 2^2 5 \cdot 11 = 220$.

Definition 5. The *lowest common multiple* (or LCM) of integers a and b is the smallest integer that is a multiple of both a and b .

Example 7. $10 = 5 \cdot 2$ and $6 = 2 \cdot 3$ so $\text{lcm}(10, 6) = 5 \cdot 2 \cdot 3 = 60$. Similarly the $\text{lcm}(1100, 440) = 2^3 5^2 11 = 2200$.

Exercise 4. Determine the following:

1. $\gcd(2^4 3^2 5 \cdot 7^2, 2 \cdot 3^3 7 \cdot 11)$
2. The lowest common multiple: $\text{lcm}(2^4 3^2 5 \cdot 7^2, 2 \cdot 3^3 7 \cdot 11)$.

Exercise 5. ★ Suppose that $a \neq 0$, $b \neq 0$, and $\text{lcm}(a, b) = m$. Prove that if n is a common multiple of a and b then $m|n$.

Exercise 6. ★ Let $a \neq 0$, $b \neq 0$, and $d = \gcd(a, b)$. Prove that if d' is any common divisor of a and b then $d'|d$.

The next theorem plays an important role in our study of algebra. The proof relies on both the well ordering principle and the division algorithm.

Theorem 2. GCD is a linear combination. For any nonzero integers a and b , there exists integers s and t such that $\gcd(a, b) = as + bt$. Furthermore, $\gcd(a, b)$ is the smallest positive integer that can be written in the form $as + bt$.

- In particular, a and b are relatively prime if and only if there exists integers s and t such that $1 = as + bt$.

Proof. Let a, b be two nonzero integers. Let $S = \{am + bn : m, n \in \mathbb{Z} \text{ and } am + bn > 0\}$. Note that $S \neq \emptyset$ since if $am + bn \leq 0$ then $a(-m) + b(-n) > 0$. By the well ordering principle we can select the smallest number in S , say $d = as + bt$. We want to show that $d = \gcd(a, b)$.

Using the division algorithm, write $a = dq + r$ where $0 \leq r < d$. If $r > 0$ then $r = a - dq = a - (as + bt)q = a - asq - btq = a(1 - sq) + b(-tq) \in S$. But this contradicts the assumption that d is smallest number in S since $a(1 - sq) + b(-tq) < as + bt$. So then $r = 0$ implying that $d|a$. Similarly we can show that $d|b$. This shows that d is a common divisor of a and b .

Now we show that d is the largest common divisor. Suppose that d' is another common divisor of a and b . So then $a = d'x$ and $b = d'y$ for integers x and y . Now $d = as + bt = (d'x)s + (d'y)t = d'(xs + yt)$ which implies that $d'|d$. Therefore $d' < d$. $\square$

Example 8. $\gcd(10, 6) = 2$ and $2 = 10(2) + 6(-3)$.

Example 9. $\gcd(4, 15) = 1$. So 4 and 15 are relatively prime. $1 = 4(4) + 15(-1)$.

We will see that theorem 2 is a very useful theoretical tool. However it does not tell us how to actually find the linear combinations. Fortunately, we have the Euclidean Algorithm³.

²often abbreviated to WLOG, WOLOG or w.l.o.g.; less commonly stated as without any loss of generality or with no loss of generality. WLOG means that although a premise of the proof is being narrowed to a special case, the proof of that special case can be easily applied to every other case.

³From proposition 2 of book seven of the *Elements* written approximately c. 300 BC.

Example 10. Euclidean Algorithm. By repeatedly using the division algorithm the $\gcd(a, b)$ can always be found by producing a decreasing sequence of remainders. The algorithm is as follows:

$$\begin{array}{rcl}
 a & = & bq_1 + r_1 & 0 < r_1 < b \\
 b & = & r_1q_2 + r_2 & 0 < r_2 < r_1 \\
 r_1 & = & r_2q_3 + r_3 & 0 < r_3 < r_2 \\
 & \vdots & & \\
 r_{n-3} & = & r_{n-1}q_{n-1} + r_{n-1} & 0 < r_{n-1} < r_{n-2} \\
 r_{n-2} & = & r_{n-1}q_n + r_n & 0 < r_n < r_{n-1} \\
 r_{n-1} & = & r_nq_{n+1} + 0 &
 \end{array}$$

By the well ordering principle, this algorithm will eventually result in a remainder of 0. Then the last nonzero remainder, r_n , is the greatest common divisor.

For example, to find the $\gcd(2520, 154)$:

$$\begin{array}{rcl}
 2520 & = & 154(16) + 56 \\
 154 & = & 56(2) + 42 \\
 56 & = & 42(1) + 14 \\
 42 & = & 14(3) + 0
 \end{array}$$

So $\gcd(2520, 154) = 14$. We can also use this algorithm to find a linear combination of 2520 and 154 that sums to 14. Using the above results, we rewrite the remainders as follows:

$$\begin{array}{rclcl}
 56 & = & 42(1) + 14 & \Rightarrow & 14 = 56 - 42(1) \\
 154 & = & 56(2) + 42 & \Rightarrow & 14 = 56 - [154 - 56(2)] \\
 & & & & 14 = (3)56 - 154 \\
 2520 & = & 154(16) + 56 & \Rightarrow & 14 = (3)[2520 - 154(16)] - 154 \\
 & & & & 14 = 2520(3) + 154(-49)
 \end{array}$$

Exercise 7. Find integers s and t so that $1 = 7 \cdot s + 11 \cdot t$. Show that s and t are not unique.

Exercise 8. Let $d = \gcd(a, b)$. If $a = da'$ and $b = db'$, show that $\gcd(a', b') = 1$.

Exercise 9. ★ If $c^2 = ab$ and $\gcd(a, b) = 1$, prove that a and b are perfect squares (Hint: By the theorem above, a and b can be written as the product of primes. They are perfect squares if and only if each prime in this prime factorization has an even power).

Lemma 1. Euclid's Lemma. Let p be a prime and a and b be integers. If $p|ab$ then $p|a$ or $p|b$.⁴

Proof. Suppose that $p|ab$ but $p \nmid a$. We want to show that $p|b$. Since $p \nmid a$, $\gcd(p, a) = 1$ and by theorem 2 we can write $1 = sp + ta$. Multiplying both sides by b we have $b = bsp + bta$. Since $p|a$, $a = pn$ for some integer n , and

$$\begin{array}{rcl}
 b & = & bsp + bta \\
 \Rightarrow b & = & bsp + btpn \\
 \Rightarrow b & = & p(bs + btn)
 \end{array}$$

so $p|b$ as needed. □

- Euclid's lemma is not necessarily true when applied to non-primes. For example, $10|25 \cdot 2$ but $10 \nmid 25$ and $10 \nmid 2$.

1.2 Modular Arithmetic

Modular arithmetic is a system of arithmetic where numbers “wrap around” or “reset” after reaching a set value called the *modulus*. You have probably been using modulus arithmetic all your life without giving it much thought. One common use is a 12-hour clock.

If the time is 5:00 then 11 hours later we do not say it is 16:00 (unless you are in the military). Time “wraps” around every 12 hours -modulus 12. Using the division algorithm, $16 = 12(1) + 4$, and we have a remainder of 4. 16 “wraps” 4 hours past 12. We write this as

$$16 \equiv 4 \pmod{12}$$

⁴This lemma first appeared as proposition 30 in book seven of Euclid's *Elements*.

read “16 is congruent/equivalent to 4 mod 12”. Adding 12 hours to 16 will wrap around to the same number, 4, as will adding 24, 36, 48 etc.

$$\begin{aligned} 4 &\equiv 4 \pmod{12} \\ 16 &\equiv 4 \pmod{12} \\ 28 &\equiv 4 \pmod{12} \\ 40 &\equiv 4 \pmod{12} \\ &\vdots \\ 4 + 12(n) &\equiv 4 \pmod{12} \end{aligned}$$

So 4, 16, 28, etc. are all members of an infinite set we call a *class*, denoted $[4]$.

Definition 6. Let a and n be integers with $n > 0$. The *congruence class of a modulo n* (denoted $[a]$) is the set of integers that are congruent to a modulo n , i.e.,

$$[a] = \{b \mid b \in \mathbb{Z} \text{ and } b \equiv a \pmod{n}\}$$

Example 11. The congruence class of 3 modulo 5 is $[3] = \{\dots, -2, 3, 8, 13, \dots\}$.

There are several useful ways to define $a \equiv b \pmod{n}$. However when first introduced to the subject it is often easiest to think of modular arithmetic in terms of the remainder.

- If $a = qn + r$ where q is the *quotient* and r is the *remainder* upon dividing a by the *modulus* n , then $a \equiv r \pmod{n}$.

Example 12. If 20 is divided by 3 then the remainder is 2. So $20 \equiv 2 \pmod{3}$ since $20 = (6)3 + 2$.

Of course if $20 \equiv 2 \pmod{3}$ then one would think that $2 \equiv 20 \pmod{3}$ (as congruence/equivalence should be an equivalence relation), however dividing 3 by 20 is not permissible as any remainder must be greater than 0 but less than 20. For this reason we define $a \equiv b \pmod{n}$ as follows:

Definition 7. $a \equiv b \pmod{n}$ if and only if $n \mid (a - b)$.

Example 13. Using the definition above, $20 \equiv 2 \pmod{3}$ since $3 \mid (20 - 2)$ and $3 \equiv 20 \pmod{3}$ since $3 \mid (2 - 20)$.

Example 14. If $a = qn + r$ then $r = a - qn \Rightarrow qn = a - r$. So then $n \mid a - r$ and $a \equiv r \pmod{n}$.

Example 15. Examples of modular arithmetic:

$$\begin{aligned} 7 &\equiv 7 \pmod{13} && \text{since } 7 = (0)13 + 7 \\ 20 &\equiv 7 \pmod{13} && \text{since } 20 = (1)13 + 7 \\ 7 &\equiv 20 \pmod{13} && \text{since } 7 = (-1)13 + 7 \\ 48 &\equiv 0 \pmod{12} && \text{since } 48 = (4)12 + 0 \\ 75 &\equiv 25 \pmod{50} && \text{since } 75 = (1)50 + 25 \end{aligned}$$

Proposition 1. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ for integers a, b, c, d, n ⁵ then

1. $a + c \equiv (b + d) \pmod{n}$
2. $ac \equiv bd \pmod{n}$

Proof. $a \equiv b \pmod{n} \Rightarrow n \mid (a - b) \Rightarrow a - b = k_1n$ and $c \equiv d \pmod{n} \Rightarrow n \mid (c - d) \Rightarrow c - d = k_2n$ for some integers k_1 and k_2 .

Part 1. $(a + c) - (b + d) = a + c - b - d = (a - b) + (c - d) = k_1n + k_2n = n(k_1 + k_2)$. Thus $n \mid [(a + c) - (b + d)]$ which implies that $a + c \equiv (b + d) \pmod{n}$.

Part 2. $a - b = k_1n$ and $c - d = k_2n$ implies that $a = k_1n + b$ and $c = k_2n + d$.

$$\begin{aligned} ac &= (k_1n + b)(k_2n + d) \\ ac &= k_1k_2n^2 + k_1dn + k_2bn + bd \\ ac &= bd + n(k_1k_2n + k_1d + k_2b) \\ ac - bd &= n(k_1k_2n + k_1d + k_2b) \end{aligned}$$

So then $n \mid (ac - bd)$ and $ac \equiv bd \pmod{n}$. □

Exercise 10. Determine $51 \pmod{13}$, $342 \pmod{85}$, and $(82 \cdot 73) \pmod{7}$.

Exercise 11. Let n be a positive integer greater than 1 and let a and b be positive integers. Prove that $a \pmod{n} \equiv b \pmod{n}$ if and only if $a \equiv b \pmod{n}$.

⁵This assumption is only necessary for part 2. While part 1 is true for all real numbers part may fail for non-integers.

1.3 Mathematical Induction

Example 16. Prove that $\sum_{k=1}^n (2k+1) = n^2$. That is, prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$.

Proof. (step 1: base case) Suppose that $n = 1$. Then $1 = 1^2 = 1$. So then the statement is true for $n = 1$.

(step 2: inductive step) Now *assume that the statement is true for some $n \geq 1$* . We want to show that it is then true for

$n+1$. If it is true for some $n \geq 1$ then $\sum_{k=1}^{n+1} (2k+1) = \sum_{k=1}^n (2k+1) + 2(n+1) - 1$ (note how we broke it down into a part we knew and the next step for $n+1$).

$$\begin{aligned} \sum_{k=1}^{n+1} (2k+1) &= \sum_{k=1}^n (2k+1) + 2(n+1) - 1 = \underbrace{\sum_{k=1}^n (2k+1)}_{=n^2 \text{ by the assumption}} + \underbrace{2(n+1) - 1}_{=2n+1} \\ &= n^2 + 2n + 1 \\ &= (n+1)^2 \end{aligned}$$

so the statement is true by induction. □

Once we have shown it is true for $n = 1$ in step 1 then we can use step 2 to show that is true for $n = 2$. Once we know it is true for $n = 2$ we know it is true for $n = 3$ by step 3 again. Continuing like this we can show it is true for any n . This process is called mathematical induction or just proof by induction.

Theorem 3. The Principle of Mathematical Induction. *Let $S(n)$ be a statement involving a variable n , and the following hold:*

1. $S(n_0)$ is true.
2. If $S(n)$ is true for some positive integer $n \geq n_0$ then $S(n+1)$ is true.

Then $S(n)$ is true for all positive integers n .

Proof. Suppose $S(b)$ is true, and $S(n+1)$ is true whenever $S(n)$ is true, but there is some n for which $S(n)$ is not true. By the well ordering principle there is a least number, say k , such that $S(k)$ is not true. $k \neq b$ by assumption, but since k is the smallest number for which S is not true $S(k-1)$ is true. This is a contradiction since then $S(k)$ is true by assumption. □

Exercise 12. Prove that $\sum_{k=1}^n 2k = n^2 + n$.

Exercise 13. Prove that $23^n - 1$ is divisible by 11 for all positive integers n .

Exercise 14. ★ The Fibonacci numbers are :1, 1, 2, 3, 5, 8, 13, 21, 34, ... In general, the Fibonacci numbers are defined by $f_1 = 1$, $f_2 = 1$, and $f_{n+2} = f_{n+1} + f_n$ for $n = 1, 2, 3, \dots$. Prove that the n^{th} Fibonacci number $f_n < 2^n$.

1.4 Additional problems

Exercise 15. ★

Let $a \neq 0$, $b \neq 0$, and $d = \gcd(a, b)$. Prove that if d' is any common divisor of a and b then $d' | d$.

Theorem 4. The Fundamental Theorem of Arithmetic. *Every integer greater than 1 is either a prime or the unique product of primes.*

Exercise 16. ★ If $c^2 = ab$ and $\gcd(a, b) = 1$, prove that a and b are perfect squares (Hint: By the theorem above, a and b can be written as the product of primes. They are perfect squares iff. each prime in this prime factorization has an even power).

The set of all congruence classes modulo n is denoted $\mathbb{Z}_n$ (read “ $Z \bmod n$ ”).

$|\mathbb{Z}_n| = n$.

Exercise 17. ★ Suppose that $a \neq 0$, $b \neq 0$, and $\text{lcm}(a, b) = m$. Prove that if n is a common multiple of a and b then $m|n$.

Exercise 18 (☺). Two sets are said to be *equal* if there exists a bijective function between them; we write $|A| = |B|$. B is said to have *strictly greater cardinality* than A if there is a one-to-one function from A into B , but there does not exist a bijective function from A into B ; we write $|A| < |B|$. Prove that $|\mathbb{N}| < |\mathbb{R}|$.